VISHNU KUMAR

3rd-year B.Tech CSE student | Full-stack and machine learning projects | Seeking ML / AI internship
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EDUCATION

Pimpri Chinchwad University, Pune *Aug 2024 - Present*
B.Tech, Computer Science and Engineering | CGPA: 7.95

PROJECTS

FastAPI Docs RAG Chatbot with Evaluation Harness |

- Built a RAG question-answering system over the official FastAPI documentation using hybrid retrieval (ChromaDB vector search with all-MiniLM-L6-v2 embeddings + BM25, fused with reciprocal rank fusion) and cross-encoder reranking
- Generated cited answers with a Groq-hosted LLM, adding disk caching, retry with backoff, and source-citation validation
- Built an evaluation harness with 42 questions (36 answerable, 6 unanswerable) measuring hit@k, MRR, LLM-judged faithfulness and correctness, and refusal accuracy
- Shipped a CLI and Streamlit UI with retrieval-mode controls and source excerpts; added YAML configuration, pinned dependencies, and 9 pytest tests

Technologies / Tools Used: *Python, PyTorch, sentence-transformers, ChromaDB, BM25, Groq API, Streamlit, pytest*

AgroFarm — Smart Agricultural Supply Chain Platform | Live Demo

- Built a full-stack platform tracking produce from farm to retailer, with role-based dashboards for farmers, collection teams, warehouses, transport, and retailers
- Implemented live shipment and vehicle tracking with GPS simulation, real-time telemetry, and notifications
- Developed inventory and warehouse allocation management with demand forecasting and stock recommendations
- Added emergency handling with nearby facility alerts and offline-friendly frontend behavior

Technologies / Tools Used: *React, TypeScript, FastAPI, SQLAlchemy, PostgreSQL/SQLite*

ResiliAI — Disaster Risk Prediction Dashboard | Live Demo

- Built an AI-driven disaster risk prediction platform for hyperlocal early warning and community resilience planning
- Integrated real-time weather, local vulnerability factors, and incident reports to forecast hazards across at-risk regions
- Delivered data visualizations that support emergency-response decisions

Technologies / Tools Used: *Next.js, TypeScript, Firebase, Data Visualization Tools*

Housing Price Prediction Model *Oct 2024 - Nov 2024*

- Built an end-to-end ML pipeline to predict house prices from a 100,000+ property dataset
- Trained Linear Regression and Random Forest models after data cleaning and feature engineering (square footage, bedrooms, bathrooms, year built, garage)
- Used an 80-20 train-test split and evaluated with R^2, RMSE, and MAE; best model reached $R^2 = 0.825$

Technologies / Tools Used: *Python, Scikit-learn, Pandas, NumPy, Matplotlib*

ACHIEVEMENTS

- **Green Quest 2026, Pimpri Chinchwad University:** First Prize in the Eco Tech (software) track of the "Best from Waste" sustainability competition for AgroFarm, a smart agricultural supply chain platform (Sep 2026)
- **CREST CTF (National-level):** Ranked 4th in the offline round of a national Capture The Flag competition hosted by Pimpri Chinchwad University (2026)
- **Hack2Ignite 2026 (National Hackathon):** Qualified for the on-campus Grand Finale as one of the top 30 teams from the 48-hour national online qualifier, as Team Agro, hosted by G H Raisoni, Pune (2026)

SKILLS

Programming Languages: C, C++, Python, Java, TypeScript

Frameworks & Libraries: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, React, Next.js, Node.js, FastAPI, SQLAlchemy, Tailwind CSS

Databases: MySQL, PostgreSQL, SQLite, Firebase

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Postman